MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSEBOTA00 - APPLIED PHYCOLOGY AND MYCOLOGY
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer : Answer any **10** questions

Each question carries **2** marks

1. **Name the principal xanthophyll in golden-brown algae** **[K] / [CO1]**

   Fucoxanthin

2. **Distinguish between micro and mesophytoplankton** **[U] / [CO2]**

   The linear dimension of Microphytoplankton is 20-200µm, whereas in mesophytoplankton is >2 mm

3. **Give any two ecological impacts of HAB** **[U] / [CO3]**
   a. Depletes dissolved Oxygen in waterbodies
   b. Releasing toxic materials harm to aquatic ecosystem
   c. Blocking sunlight
   d. Habitat destruction

4. **Does growing algae in large "open ponds" cost less than using high-tech "closed photobioreactors"?** **[U] / [CO2]**

   Yes, growing algae in large open ponds (often called raceway ponds) generally costs less than using high-tech closed photobioreactors (PBRs) . Open ponds are recognized for having minimal capital construction costs, lower operating costs, and reduced energy requirements for mixing.

5. **Define soxlet extraction?** **[U] / [CO2]**

   Soxhlet extraction in algae extraction is a continuous, automated solvent-based technique used to efficiently isolate total lipids and oils from dried algal biomass .

6. **Name any two food items produced from algae** **[K] / [CO2]**

   Nori, kombu, Wakame, Dulse

7. **What is Agar Agar?** **[U] / [CO3]**

   Agar-agar is a gelatinous substance extracted from red algae (species like *Gelidium* and *Gracilaria*), used extensively as a solidifying agent for plant tissue culture media.

8. Name the major classes of fungal enzymes used in industry. [K] / [CO4]

Hydrolases – Amylases,Protease ,lipaeses (Any one)

9. Which mushroom is commonly cultivated for food security? [K] / [CO4]

Agaricus bisporus or button mushroom

10. Which organ is primarily affected by afflatoxin exposure. [U] / [CO4]

Liver

11. Name one pharmaceutical compound derived from lichen. [U] / [CO4]

Usnic acid from Usnea and Protolichesterinic acid from Cetraria

12. Give an example of bioactive compounds produced by fungal endophyte [U]/[CO4]

Alkaloid, Flavonoids, terpenoids

Examples: Taxol from Taxomyces, Podophyllotoxin from Phialocephala,

Part B

Short Essay Answer any **8** questions
Each question carries **5** marks

13. List the types of ribosomes and their rRNA composition as used in algal systematics.

[K]/[CO1]

1) Prokaryotic ribosomes. The large 70S subunit contains 5S and 23S rRNAs as well as 34 ribosomal proteins. The small 30S subunit contains a single 16S rRNA and 21 proteins.
2) Eukaryotic ribosomes. The large 60S subunit contains 28S, 5.8S, and 5S rRNAs, and 49 proteins. The small 40S subunit contains 18S RNA and 33 proteins.
3) Mitochondrial ribosomes. These are similar, although not the same, as prokaryotic ribosomes. They are not used to produce algal phylogenies

14. Expand ATCC and its significance [K] / [CO2]

American Type Culture Collection – ATCC, large and well-known repository of algae cultures. Cultures of many algal species are readily available from collections maintained by experts in several nations, for various experimental studies.

15. Describe fungal bio fertilizers with suitable examples. [K]/[CO3]

Fungal biofertilizers are preparations containing beneficial fungi that enhance plant growth by improving nutrient availability, uptake, and soil health. Unlike chemical fertilizers, they work through natural biological processes in the soil–plant system.

Mycorrhizal fungi- Glomus

Phosphate solubilizing fungi- *Aspergillus niger*, *Penicillium* Decomposer fungi, Tricoderma

16. Describe various applications of algal extracts. [A] / [CO2]

For producing Biofertilizers

Bioplastic, biopolymers, Biofuels like biogas, biodiesel, bioethanol and biohydrogen

Food, feed and supplements

Nutraceuticals and therapeutics

Pharmaceuticals and cosmetics formulations

17. Explain different molecular genetic techniques for Algal bioengineering. [U] / [CO2]

Molecular genetic techniques are concerned with manipulating, reproducing, adding, and deleting DNA and RNA molecules in an organism.

Manipulations, deletions, and additions are accomplished via genetic transformation and genetic editing, while reproduction is accomplished via cloning.

Genetic transformation in algae involves introducing foreign DNA into nuclear or chloroplast genomes using methods like electroporation, particle bombardment, Agrobacterium-mediated transfer, and glass bead agitation to alter metabolic pathways.

18. Why is yeast prefered in the baking industry? [U] / [CO3]

Yeast is preferred in the baking industry mainly because of how effectively it improves both the texture and flavor of baked goods.

At its core, yeast works through a process called Fermentation. When added to dough, yeast feeds on sugars and produces carbon dioxide gas and a small amount of alcohol. The gas gets trapped in the dough, causing it to rise and become light and airy—this is what gives bread its soft, fluffy structure.

19. Explain how fungi serve as model organisms in scientific research. [U] / [CO4]

- Easy to culture: They grow quickly on simple, inexpensive media.
- Short life cycles: This allows rapid observation of multiple generations.
- Genetic simplicity: Many fungi have small, well-characterized genomes.
- Eukaryotic cells: Unlike bacteria, fungi have a nucleus and organelles, making them more comparable to animal and plant cells.
- Genetic manipulation: They are easy to mutate and study genetically.

Common fungal model organisms

Saccharomyces cerevisiae; Schizosaccharomyces pombe; Neurospora crassa; Aspergillus nidulans

20. Explain the importance of sanitation in storage management. [U] / [CO3]

Prevention of Microbial Contamination

Reduction of Pest Infestation Maintenance of Product quality Prevention of cross contamination

Compliance with Health and Safe regulations

Economic benefits-Reduces product loses due to spoilage orinfestation, Induced Resistance, avoid mechanical injuries, Ensure clean packaging and storage conditions

21. Explain the concept of nutraceuticals and discuss the role of fungi as nutraceutical source. [U] / [CO4]

Nutraceuticals are food, supplements or products that deliver medicinal or physiological benefits in addition to nutrition.

Fungi, particularly medicinal mushrooms, are considered one of the most prolific sources of **nutraceuticals**—products that provide physiological benefits or protection against chronic

diseases beyond basic nutrition. Fungi, particularly edible and medicinal mushrooms, are considered "mycopharmaceuticals" due to their dual role as food and medicine.

Polysaccharides (β-glucans): Found in mushrooms like *Lentinula edodes* and *Ganoderma lucidum*. They enhance immune response and have anti-tumor properties.Phenolic compounds & antioxidants: Help reduce oxidative stress and protect against degenerative diseases.

Triterpenoids: Present in Ganoderma, known for anti-inflammatory and hepatoprotective effects.
Ergothioneine & vitamins (D2): Mushrooms are one of the few non-animal sources of vitamin D, beneficial for bone and immune health.

Secondary metabolites: Some fungi produce statins (e.g., *Monascus purpureus* produces monacolin K, a cholesterol-lowering compound). (Any five points)

22. Explain the concept of bio-prospecting in fungi. [U] / [CO4]

Bioprospecting is the systematic search for organisms that produce valuable biochemical compounds.

The goal is to identify fungi that naturally produce volatile hydrocarbons or hydrocarbon like molecules similar to petroleum fuels.

Some fungi synthesize volatile organic compounds and long chain hydrocarbons that resemble diesel or gasoline compounds

[5x8 = 40]

Part C

Essay

Answer any **1** question; Each question carries **10** marks **[2x10 = 20]**

23. Analyze the future prospects of algal biotechnology. [U] / [CO2]

Algal biotechnology is rapidly emerging as a critical component of the future green bioeconomy, offering sustainable solutions for energy, nutrition, and environmental management. As photosynthetic organisms, algae present a versatile "cell factory" for producing high-value compounds, including biofuels, pharmaceuticals, pigments, and nutritional supplements.

- Advanced Biofuels & Renewable Energy: Algae are considered the ultimate source for third-generation biofuels, including biodiesel, bioethanol, and biohydrogen. Future prospects lie in increasing lipid productivity and optimizing thermochemical conversion technologies to make algal fuels cost-competitive with fossil fuels.
- Pharmaceuticals and Nutraceuticals: Algae produce potent bioactive molecules, such as carotenoids (astaxanthin, -carotene) and polyunsaturated fatty acids (PUFAs), which have antioxidant and anti-inflammatory properties.
- Environmental Remediation (Phycoremediation): Algae are exceptionally effective at treating industrial and municipal wastewater, removing nitrogen, phosphorus, and heavy metals while producing usable biomass. They can also mitigate atmospheric emissions by capturing it

during photosynthesis, aiding in carbon footprint reduction.

- Bioproducts & Materials: algae are being used to develop sustainable alternatives to petroleum-based products, including bioplastics, biodegradable fabrics, and high-value pigments.
- Genetic Engineering & Synthetic Biology: The application of CRISPR-Cas9 and other genome-editing tools allows researchers to modify algal genomes for higher yields, resilience to harsh conditions, and optimized nutrient synthesis

24. Explain how fungi help in the breakdown of agro-residues. [U] / [CO4]

Fungi play a crucial role in breaking down agro-residues (like. straw, husk, stalks, and leaves left after harvest) by acting as nature's primary recyclers. They decompose complex plant materials and convert them into simpler substances that enrich the soil.

- Breaking down tough plant components
  Agro-residues are rich in complex compounds such as: Cellulose,Hemicellulose, lignin
  These are difficult for most organisms to digest. Fungi are. uniquely equipped to degrade them using powerful enzymes:
  Cellulases break cellulose into simple sugars Hemicellulases break hemicelluloseLigninases (like laccase and peroxidase) break down lignin, which is the hardest component
  Turning residues into simpler organic matter As fungi grow on crop waste, they:
- Absorb nutrients from the residues
  Chemically break them into simpler organic compounds Convert them into humus-like material
- Improving soil fertility
  The breakdown process helps:
  Release nutrients like nitrogen, phosphorus, and potassium back into the soil Increase soil organic matter
- Speeding up composting
  Fungi are especially important in composting agricultural waste:
  They colonize residues early and soften tough plant fibers Work alongside bacteria and other microbes
- Help faster and more efficient compost formation

  Examples of useful fungi
  Some fungi commonly involved include:
  White-rot fungi (efficient lignin degraders) Brown-rot fungi (mainly cellulose breakdown)
  Species like Trichoderma and Pleurotus (oyster mushrooms), which are widely used in agricultural waste decomposition

[10x1 = 10]